

A Disproof of Conjecture 00000004607: Convexity, Nonnegativity, and Two Endpoint Zeros

Independent solution

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Abstract

The numerical properties asserted for the concavity correction are mutually inconsistent on every volume interval $[0, V]$ with $0 < V < \infty$. A convex nonnegative function with zero values at the two endpoints must vanish throughout the interval, whereas the conjecture requires that only the endpoints be zeros. This obstruction uses neither symmetry nor a formula for the correction. The proof is formalized with Mathlib’s ordinary real convexity.

1 What the original statement asserts

The English statement defines the isoperimetric profile as the minimal interface area needed to enclose a given volume. It then asserts that its concavity correction is an explicit nonnegative function of volume, “vanishing only at zero and at the total volume,” and that the function is convex and symmetric. The Chinese statement asserts the same properties; its zero-set clause specifically refers to zero volume and total volume as the two points where the function is zero. The parenthetical descriptions in both languages also refer to two zeros. The complete, unaltered bilingual statement is reproduced in `provenance/ORIGINAL.md`.

Consequently, if $V > 0$ is the finite total volume and c is the asserted real-valued function, the following are necessary properties:

$$c \text{ is convex on } [0, V], \tag{1}$$

$$c(v) \geq 0 \quad (0 \leq v \leq V), \tag{2}$$

$$c(v) = 0 \iff v = 0 \text{ or } v = V \quad (0 \leq v \leq V). \tag{3}$$

In particular, both endpoint values are zero. The usual volume-complement symmetry would be $c(V - v) = c(v)$. We include this in the final formal statement, but the contradiction does not depend on how symmetry is interpreted, because (1)–(3) alone are impossible.

Neither language defines the operation called a concavity correction. We do not supply a definition of that operation, identify it with a particular deficit, or assume a relation between it and a chosen profile. Instead, we disprove necessary properties that the text itself explicitly imposes on its output. Any proposed definition satisfying the original conclusion would have to produce a function satisfying (1)–(3).

2 The obstruction

Theorem 1. *Let $V \geq 0$ be a real number. If $c : [0, V] \rightarrow \mathbb{R}$ is convex and nonnegative and $c(0) = c(V) = 0$, then $c(v) = 0$ for every $v \in [0, V]$.*

Proof. If $V = 0$, the interval has only its zero endpoint. If $V > 0$, fix $v \in [0, V]$ and set $t = v/V \in [0, 1]$. Then $v = (1 - t)0 + tV$, so convexity gives

$$c(v) \leq (1 - t)c(0) + tc(V) = 0.$$

Nonnegativity gives the reverse inequality, hence $c(v) = 0$. □

Corollary 1. *For every real $V > 0$, no real-valued function on $[0, V]$ satisfies (1)–(3). In particular, no explicit symmetric function satisfies all of the properties in the conjecture.*

Proof. Theorem 1 implies $c(V/2) = 0$. But $0 < V/2 < V$, contradicting (3). Requiring a formula or symmetry cannot restore existence. □

Equivalently, just the midpoint case of convexity already gives

$$0 \leq c(V/2) \leq \frac{1}{2}c(0) + \frac{1}{2}c(V) = 0,$$

while the prescribed zero set and nonnegativity require $c(V/2) > 0$. There are no smoothness, continuity, differentiability, or strict-convexity assumptions in this argument.

3 Why the volume interval is a legitimate setting

Finite positive total volume is the ordinary nondegenerate interpretation of the text’s two endpoint volumes. It is not a curvature hypothesis or a geometric restriction introduced to obtain a special formula. For example, the flat square torus $(\mathbb{R}/\mathbb{Z})^2$, with its standard product area measure, has total area 1. For every $v \in [0, 1]$, the image of $[0, v] \times [0, 1]$ has area v . For $0 < v < 1$ it is a strip bounded by two circles of finite total length 2; at the endpoints one uses the empty set and the whole torus. Thus every volume in $[0, 1]$ is admissible in a standard compact, connected, smooth isoperimetric setting. In the usual infimum formulation of the profile, these competitors make its value finite and nonnegative. No computation of the profile or existence theorem for minimizers is used in the disproof.

More abstractly, if the domain of volumes is convex and contains 0 and V , it necessarily contains the whole interval between them, in particular $V/2$. This is also built into Mathlib’s `ConvexOn` predicate. The claimed convexity therefore cannot evade the midpoint merely by discarding interior volumes. Restricting the domain to the two endpoints would change the usual meaning of convexity and would not describe the example above.

When $V = 0$, the two endpoints coincide and the identically zero function on the singleton interval presents no contradiction. This degenerate case cannot establish a claim about two distinct zeros in positive volume. Infinite total volume is not a second real endpoint, and the formula $V - v$ is then not a real-valued reflection of a bounded interval. The original text supplies no alternative definitions for that case. Our disproof concerns every finite positive total volume, which includes the concrete ordinary setting just described. It makes no claim that all conceivable isoperimetric spaces have finite volume. In particular, no infinite-volume extension can rescue the asserted conclusion in this finite-volume setting.

4 Lean formalization and its boundary

The project is pinned to Lean 4.19.0 and Mathlib commit

c44e0c8ee63ca166450922a373c7409c5d26b00b.

`Disproof.lean` contains these theorems in the namespace `IsoperimetricCorrection`:

- `zero_on_volume_interval`: Theorem 1, for every real $V \geq 0$.
- `no_nonnegative_convex_double_zero`:
The nonexistence in Corollary 1, for every real $V > 0$.
- `no_shape_requirements`: nonexistence with the additional condition $c(V - v) = c(v)$, again for every real $V > 0$.

The definition `ShapeRequirements` groups these explicitly stated numerical properties; it is not a definition of a concavity correction. The function is represented as $f : \mathbb{R} \rightarrow \mathbb{R}$, with every hypothesis restricted to $[0, V]$. Values outside the interval are immaterial; any function defined only on the interval can be extended there arbitrarily.

The formal proof uses the stock predicates `ConvexOn` and `Set.Icc`. It obtains the upper bound on each value from `ConvexOn.le_on_segment` and `Icc_subset_segment`, combines it with nonnegativity, and applies the exact zero-set condition at $V/2$. The only positivity hypothesis on the total volume in the final theorems is $0 < V$. No hypothesis asserts the desired conclusion or a geometric theorem that has not been established.

The Lean project does not formalize the geometry of the torus, the isoperimetric profile, a specific correction operator, or an interpretation of “explicit.” These are unnecessary for the universal scalar obstruction; the torus paragraph only verifies in ordinary mathematics that the volume domain has a familiar geometric realization. The logical passage from the bilingual prose to the necessary scalar properties is explained above, rather than hidden as an extra axiom.

5 Verification and provenance

The entire project was built against pre-existing stock Mathlib dependencies; this is not a claim that all of Mathlib was rebuilt. Each source file was also replayed with `warningAsError=true`. The axiom audit of all three final theorems reports only `propext`, `Classical.choice`, and `Quot.sound`. There are no admitted proofs, custom axioms, or uses of `native_decide`. Build outputs, exit codes, and the unsuccessful development attempts are preserved under `verification/` and `provenance/`, as described in `README.md`.

The author input was restricted to the original bilingual statement, both competition rule files, and stock Lean configuration. The statement’s SHA-256 is recorded in `provenance/INPUTS.json`. No prior solution, public claim history, selector material, or other author project was used as mathematical input. The mathematical argument is the elementary endpoint inequality above; there are no auxiliary numerical computations or external mathematical sources.